SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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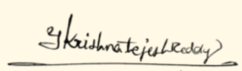
Date:	29-7-2026	**Duration:**	60 minutes

Code of Conduct

I _Y. Krishna Tejesh Reddy (192411200) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



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Annexure A – Case Study Analysis

1 Understand the Problem Statement

1.1 Explain the problem in your own words.

The Campus Placement and Recruitment Management Portal is a web-based software system developed to automate and simplify the campus recruitment process. In many educational institutions, placement activities such as maintaining student records, sharing company notifications, collecting resumes, scheduling interviews, and publishing results are managed manually using spreadsheets, emails, and paper documents. These traditional methods are time-consuming, prone to errors, and difficult to manage when the number of students and recruiters increases. The proposed portal provides a centralized platform where students, placement officers, recruiters, and administrators can efficiently manage all placement activities. By digitizing the recruitment process, the system improves communication, reduces paperwork, ensures data accuracy, and enhances the overall placement experience.

1.2 Identify the objectives of the proposed system.

- Automate the complete campus placement and recruitment process.
- Maintain student, recruiter, and company information in a centralized database.
- Enable recruiters to post job opportunities online.
- Allow students to apply for eligible job openings through the portal.
- Simplify interview scheduling and result publication.
- Provide real-time notifications for placement activities.
- Generate placement reports and recruitment statistics automatically.
- Improve transparency, efficiency, and communication among all stakeholders.

1.3 State the scope of the problem.

The scope of the Campus Placement and Recruitment Management Portal includes managing all activities related to campus recruitment within an educational institution. The system supports student registration, profile management, resume uploads, company registration, job posting, online applications, interview scheduling, and result publication through a single integrated platform. It also provides role-based access for students, placement officers, recruiters, and administrators to ensure secure and organized management of placement activities. Additionally, the portal generates placement reports and maintains a centralized database for future reference. However, the system is limited to campus recruitment activities and does not include post-recruitment HR functions such as payroll, employee attendance, or performance management.

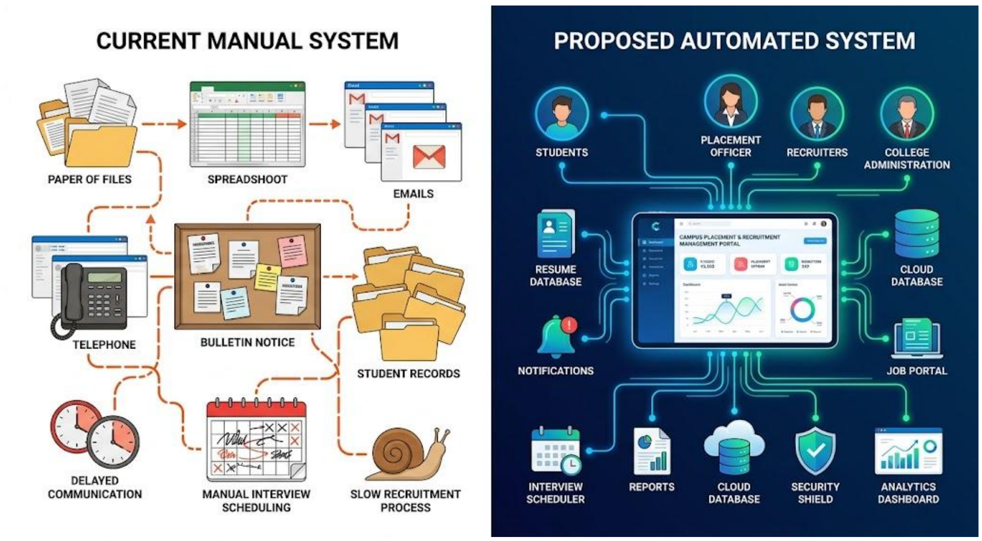

Fig 1 : Current vs Proposed Campus Placement System

2 Identify Key Stakeholders and Challenges

2.1 Identify all the stakeholders involved in the system.

- ✓ **Students** – Register on the portal, maintain profiles, upload resumes, apply for jobs, and track application status.
- ✓ **Placement Officers** – Manage placement drives, verify student eligibility, coordinate with recruiters, and generate placement reports.
- ✓ **Recruiters/Companies** – Register their organizations, post job vacancies, shortlist candidates, schedule interviews, and publish results.
- ✓ **College Administration** – Monitor placement performance, review reports, and oversee placement activities.
- ✓ **System Administrator** – Maintain the software, manage user accounts, ensure data security, and perform system updates.

2.2 Explain the roles and responsibilities of each stakeholder.

Each stakeholder plays a vital role in ensuring the smooth functioning of the Campus Placement and Recruitment Management Portal. Students are responsible for maintaining updated academic profiles, uploading resumes, and applying for suitable job opportunities. Placement officers verify student information, organize placement drives, communicate with recruiters, and monitor the recruitment process. Recruiters use the portal to publish job openings, define eligibility criteria, review applications, shortlist candidates, conduct interviews, and announce final selections. The college administration supervises overall placement performance and evaluates recruitment statistics. The system administrator is responsible for maintaining the portal, ensuring database security, managing user accounts, and providing technical support whenever required.

2.3 Discuss the major challenges and issues faced by the stakeholders.

- Manual maintenance of student and company records consumes significant time.
- Students may miss important notifications regarding placement drives and interviews.
- Recruiters find it difficult to identify eligible candidates quickly.
- Placement officers spend considerable effort verifying student eligibility manually.
- Poor communication delays recruitment activities.
- Manual report generation increases the chances of human errors.
- Lack of a centralized database creates duplicate and inconsistent records.
- Security risks may arise due to improper management of confidential information.

3 Analyze the Current Approach

3.1 Describe the existing system or current process.

The existing campus placement process in many educational institutions is managed manually using spreadsheets, emails, notice boards, and paper documents. Student records, company details, and placement information are maintained separately, making data management difficult and time-consuming. Placement officers manually verify student eligibility, communicate with recruiters, schedule interviews, and publish placement results. Students depend on emails or notice boards for updates, which can lead to delays or missed opportunities. As the number of students and recruiting companies increases, the manual system becomes less efficient, resulting in poor coordination and slower recruitment activities.

3.2 Identify its strengths and limitations.

- ❖ Easy to understand and implement.
- ❖ Requires minimal technical infrastructure.
- ❖ Low initial implementation cost.
- ❖ Familiar process for staff and students.
- ❖ Manual data entry increases the chances of human errors.
- ❖ Placement records are difficult to maintain and update.
- ❖ Communication through emails and notice boards is slow.

3.3 Analyze the problems or gaps that need to be addressed.

The existing placement process has several shortcomings that affect its efficiency and reliability. The absence of a centralized database makes it difficult to maintain accurate and updated student records. Manual communication often results in delayed notifications and missed placement opportunities. There is no automated system for verifying eligibility, scheduling interviews, or tracking application status, leading to unnecessary delays and confusion. In addition, limited security measures and the possibility of human errors reduce the reliability of the system. These gaps highlight the need for a software-based solution that automates placement activities, improves communication, enhances data security, and provides better transparency throughout the recruitment process.

4 Propose a Suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem.

The proposed solution is a web-based **Campus Placement and Recruitment Management Portal** that automates and manages the entire campus recruitment process through a centralized platform. The portal allows students, placement officers, recruiters, and administrators to access placement-related services according to their roles. Students can create profiles, upload resumes, and apply for jobs, while recruiters can post vacancies, shortlist candidates, and schedule interviews. Placement officers can manage recruitment drives, verify eligibility, and monitor placement activities efficiently. The system also provides secure data storage, real-time notifications, and automated report generation, making the recruitment process faster, more transparent, and highly organized.

4.2 Describe the major functional modules of the proposed system.

Student Module

- Student registration and secure login.
- Profile creation and academic record management.
- Resume upload and update.
- Search and apply for job opportunities.
- View interview schedules and placement results.

Recruiter Module

- Company registration and authentication.
- Post job vacancies and eligibility criteria.
- Review and shortlist student applications.
- Schedule interviews and publish final results.
- Track recruitment progress.

Placement Officer Module

- Verify student eligibility.
- Manage placement drives.
- Coordinate with recruiters.
- Monitor placement activities.
- Generate placement reports.

Administrator Module

- Manage users and access permissions.
- Maintain the system database.

- Monitor system security.
- Perform backups and software updates.
- Generate overall system reports.

4.3 Identify the functional and non-functional requirements

Functional Requirements

- ✓ User registration and secure login.
- ✓ Student profile management.
- ✓ Resume upload facility.
- ✓ Company registration and job posting.
- ✓ Online job applications.
- ✓ Eligibility verification.
- ✓ Interview scheduling.
- ✓ Notification system.
- ✓ Placement report generation.
- ✓ Search and filter options.

Non-Functional Requirements

- High system performance.
- Reliable and accurate operation.
- Secure authentication and data protection.
- Scalable architecture.
- User-friendly interface.

4.4 Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

The proposed Campus Placement and Recruitment Management Portal is developed by applying important software engineering principles to ensure quality, efficiency, and long-term maintainability. The principle of **modularity** divides the application into independent modules such as Student, Recruiter, Placement Officer, and Administrator, making development and maintenance easier. **Scalability** enables the system to support an increasing number of users and placement drives without affecting performance. **Maintainability** allows developers to modify or upgrade individual modules without impacting the entire application. **Reliability** ensures accurate processing of placement data, while **usability** provides a simple and user-friendly interface for all stakeholders.

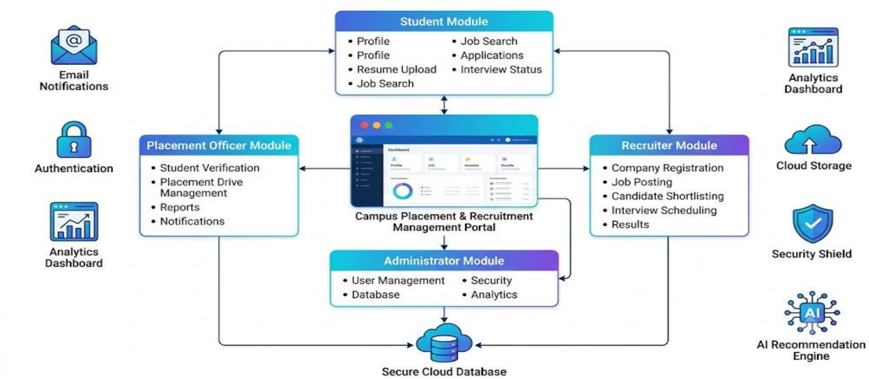

Fig 4.1 : Software Architecture Diagram

5 Justify the Chosen Methodology/Framework

5.1 Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

The **Agile Software Development Methodology**, particularly the **Scrum framework**, is selected for developing the Campus Placement and Recruitment Management Portal. Agile is an iterative and flexible approach that divides the project into smaller development cycles called sprints. This methodology allows the development team to deliver functional modules incrementally while continuously improving the system based on stakeholder feedback. Since placement requirements frequently change due to new company policies, eligibility criteria, and recruitment procedures, Agile provides the flexibility needed to accommodate these changes efficiently. It also promotes collaboration, faster development, and high-quality software delivery.

5.2 Justify why the selected methodology/framework is suitable for the proposed system.

Agile is the most suitable methodology for this project because it supports continuous improvement and adapts easily to changing requirements. The placement portal may require new features such as additional recruiter functionalities, updated eligibility criteria, AI-based recommendations, or enhanced reporting capabilities during development. Agile enables developers to implement these changes without affecting the overall project. Regular interaction with placement officers, recruiters, and students helps the development team understand user requirements more clearly and deliver a solution that meets their expectations. This flexibility makes Agile an effective choice for developing a dynamic and user-oriented placement management system.

5.3 Explain how it supports successful project development and maintenance.

Agile supports successful project development by promoting continuous testing, regular stakeholder feedback, and incremental software delivery. Each sprint produces a functional part of the system, allowing issues to be identified and corrected at an early stage. This approach improves software quality, reduces development risks, and ensures that the final application satisfies user requirements. Agile also simplifies future maintenance because enhancements and new features can be added without redesigning the entire system. Continuous collaboration among developers, testers, placement officers, and recruiters ensures better communication, faster decision-making, and efficient project management. As a result, the software remains reliable, maintainable, and adaptable to future placement requirements.

6 Discuss Expected Outcomes and Limitations

6.1 Describe the expected benefits of the proposed solution.

- Automates the complete campus placement and recruitment process.
- Reduces manual paperwork and administrative workload.
- Provides a centralized database for student and recruiter information.
- Improves communication through real-time notifications.
- Enables recruiters to identify eligible candidates quickly.
- Generates accurate placement reports and statistics.
- Enhances transparency, efficiency, and overall placement management.
- Saves time and improves the placement experience for all stakeholders.

6.2 Explain how the solution addresses the identified challenges.

The proposed Campus Placement and Recruitment Management Portal effectively addresses the challenges of the existing manual placement process by providing a centralized and automated platform. It eliminates manual record maintenance by storing all student, recruiter, and placement data in a secure database. The portal automates eligibility verification, job applications, interview scheduling, and result publication, significantly reducing delays and human errors. Real-time notifications keep students informed about placement activities, while recruiters can easily shortlist eligible candidates using search and filtering options. The system also generates reports automatically, helping placement officers and administrators monitor recruitment performance efficiently.

6.3 Discuss potential risks, implementation challenges, and limitations.

- ✓ Requires a stable internet connection for uninterrupted access.
- ✓ Initial development and implementation costs may be high.
- ✓ Users may require training to use the portal effectively.
- ✓ Server failures or downtime may temporarily affect system availability.
- ✓ Data backup and recovery mechanisms must be implemented.
- ✓ Continuous monitoring is required to maintain system performance and security.

6.4 Suggest possible future enhancements.

- Integrate AI-based job recommendation features.
- Implement automated resume screening using Machine Learning.
- Develop a mobile application for Android and iOS users.
- Add online coding tests and aptitude assessments.
- Provide advanced analytics dashboards for placement statistics.
- Integrate professional networking platforms such as LinkedIn.
- Deploy the system on cloud infrastructure for better scalability.
- Add blockchain-based certificate verification for secure credential validation.

Fig 6.1 : Future Enhancements